STAT451 – Final Exam Formula Sheet

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1. Descriptive Statistics

Sample mean: $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$

Sample variance & standard deviation:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}x_i^2 - n\bar{x}^2\right)$$

$$s = \sqrt{s^2}$$

Population mean & variance:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2$$

Median (ordered $x_{(1)} \le \cdots \le x_{(n)}$):

$$\text{Med} = \begin{cases} x_{\left(\frac{n+1}{2}\right)} & n \text{ odd} \\ \frac{1}{2}\left(x_{(n/2)} + x_{(n/2+1)}\right) & n \text{ even} \end{cases}$$

Mode: most frequent value. **Range:** $R = x_{\max} - x_{\min}$.

2. Binomial & Normal Distributions

Binomial $X \sim B(n,p)$:

$$P(X=k) = \binom{n}{k}p^k(1-p)^{n-k}, \quad k = 0,1,\dots,n$$

$$E(X) = np, \qquad \text{Var}(X) = np(1-p)$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Normal $X \sim \mathcal{N}(\mu,\sigma^2)$:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Standardization (Z-score):

$$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$$

$$P(a \le X \le b) = \Phi\left(\tfrac{b-\mu}{\sigma}\right) - \Phi\left(\tfrac{a-\mu}{\sigma}\right)$$

Note: Φ is the CDF of $\mathcal{N}(0,1)$.

$$\Phi(z) = P(Z \le z), \quad Z \sim \mathcal{N}(0,1)$$

Identities: $\Phi(-z) = 1-\Phi(z)$; $P(|Z| \le z) = 2\Phi(z) - 1$; $P(Z > z) = 1 - \Phi(z)$.

3. Sampling Distributions & CLT

Standard Error (SE): For $\bar{X}$ (sample mean diameter), $\text{SE} = \sigma/\sqrt{n}$.
Sample mean (from $\mathcal{N}(\mu,\sigma^2)$):

$$\bar{X} \sim \mathcal{N}\left(\mu, \tfrac{\sigma^2}{n}\right), \quad Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$$

Central Limit Theorem: for large n and any population with mean μ, variance σ^2,

$$\bar{X} \overset{\text{approx}}{\sim} \mathcal{N}\left(\mu, \tfrac{\sigma^2}{n}\right), \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1)$$

t-statistic (σ unknown, normal pop):

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$$

Sample proportion $\hat{p} = X/n$; for large n (rule: np, $n(1-p) \ge 5$):

$$\hat{p} \overset{\text{approx}}{\sim} \mathcal{N}\left(p, \tfrac{p(1-p)}{n}\right), \quad Z = \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \sim \mathcal{N}(0,1)$$

Normal approximation to Binomial: if $X \sim B(n,p)$, $np \ge 10$, $n(1-p) \ge 10$:

$$X \overset{\text{approx}}{\sim} \mathcal{N}(np, np(1-p))$$

With continuity correction:

$$P(X \le k) \approx \Phi\left(\tfrac{k+0.5-np}{\sqrt{np(1-p)}}\right)$$

(Calculate the area to the left. In Casio: Lower bound: $k+0.5$, Upper bound: 10.)

$$P(X \ge k) \approx 1 - \Phi\left(\tfrac{k-0.5-np}{\sqrt{np(1-p)}}\right)$$

(Calculate the area to the right. In Casio: Lower bound: -10, Upper bound: $k-0.5$.)

4. Confidence Intervals (One Sample)

Confidence level $1-\alpha$. Two-sided critical values: $z_{\alpha/2}$ (normal), $t_{n-1,\alpha/2}$.

(a) μ, σ known (any n if population normal, else large n):

$$\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ is called Margin of Error (ME) for mean when σ known.
Sample size for mean (given margin E):

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2 \quad \text{(round UP)}$$

(b) μ, σ unknown, large n ($n \ge 30$):

$$\bar{X} \pm z_{\alpha/2}\frac{S}{\sqrt{n}}$$

(c) μ, σ unknown, small n (normal population):

$$\bar{X} \pm t_{n-1,\,\alpha/2}\frac{S}{\sqrt{n}}$$

(d) Population proportion p (large n: $n\hat{p}, n(1-\hat{p}) \ge 5$):
Note: $\hat{p} = X/n$ (X=count with property; n=sample size).

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

5. Hypothesis Testing – One Sample

Steps: (1) state H_0, H_1; (2) choose α; (3) pick test statistic; (4) find rejection region or p-value; (5) decide; (6) conclude in context.
Note: p_0 is the hypothesized value under H_0.
Test for mean μ ($H_0: \mu = \mu_0$):

- σ known, or $n \ge 30$: $Z_0 = \dfrac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$
- σ unknown, large n: $Z_0 = \dfrac{\bar{X}-\mu_0}{S/\sqrt{n}}$
- σ unknown, small n, normal pop: $T_0 = \dfrac{\bar{X}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$

Test for proportion p ($H_0: p = p_0$):

$$Z_0 = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}} \sim \mathcal{N}(0,1)$$

Note: n=sample size; SE uses p_0 (not $\hat{p}$) in denominator.
Rejection regions & p-values – Z test:
Assuming z_α is positive by default

H_1	Reject H_0 if	p-value
$\mu \ne \mu_0$	$\lvert Z_0\rvert > z_{\alpha/2}$ (Two-tailed)	$2[1-\Phi(\lvert Z_0\rvert)]$
$\mu > \mu_0$	$Z_0 > z_\alpha$ (Right-tailed and positive)	$1-\Phi(Z_0)$
$\mu < \mu_0$	$Z_0 < -z_\alpha$ (Left-tailed and negative)	$\Phi(Z_0)$

For T-test replace $z \to t_{n-1}$ and $\Phi \to F_{t_{n-1}}$.
Decision: reject H_0 iff p-value $\le \alpha$.

6. Hypothesis Testing – Two Independent Samples

Samples: $(\bar{X}_1, S_1, n_1)$ vs $(\bar{X}_2, S_2, n_2)$; hypothesized Δ_0. $H_0 : \mu_1 - \mu_2 = \Delta_0$; often $\Delta_0 = 0$.

Large n_1, n_2: both $n_1 > 30$ **and** $n_2 > 30$. **Small** n: either $n_1 \le 30$ **or** $n_2 \le 30$.

(a) Both σ_1, σ_2 known:

$$Z_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim \mathcal{N}(0,1)$$

CI: $(\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2}\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$.

(b) Large n_1, n_2, σ's unknown: same form using S_1, S_2:

$$Z_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

CI: $(\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2}\sqrt{S_1^2/n_1 + S_2^2/n_2}$.

(c) Small n, $\sigma_1^2 = \sigma_2^2$ (pooled t-test):

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

$$T_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

CI: $(\bar{X}_1 - \bar{X}_2) \pm t_{n_1+n_2-2,\,\alpha/2}\, S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$.

(d) Small n, $\sigma_1^2 \neq \sigma_2^2$ (Welch's t):

$$T_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \sim t_\nu$$

$$\nu = \frac{\left(S_1^2/n_1 + S_2^2/n_2\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}} \quad \text{(round down)}$$

CI: $(\bar{X}_1 - \bar{X}_2) \pm t_{\nu,\alpha/2}\sqrt{S_1^2/n_1 + S_2^2/n_2}$.

(e) Two proportions ($H_0 : p_1 = p_2$), pooled:

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim \mathcal{N}(0,1)$$

CI for $p_1 - p_2$ (unpooled SE):

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Rejection region (two-tailed): Z-test: $|Z_0| > z_{\alpha/2}$; pooled T-test: $|T_0| > t_{n_1+n_2-2,\alpha/2}$; Welch: $|T_0| > t_{\nu,\alpha/2}$. One-tailed analogous.

7. Two Dependent Samples – Paired t-test

Differences $D_i = X_i - Y_i$, $i = 1, \dots, n$.

$$\bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \quad S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2$$

Test $H_0 : \mu_D = \mu_{D,0}$ (usually 0):

$$T_0 = \frac{\bar{D} - \mu_{D,0}}{S_D/\sqrt{n}} \sim t_{n-1}$$

Confidence interval for μ_D:

$$\bar{D} \pm t_{n-1,\,\alpha/2}\,\frac{S_D}{\sqrt{n}}$$

Reject (two-tailed) if $|T_0| > t_{n-1,\alpha/2}$. *Use when the two samples are paired (same subject, before/after, matched pairs).*

8. Simple Linear Regression

Model: $Y = \beta_0 + \beta_1 X + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

Auxiliary sums:

$$S_{xx} = \sum(x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

$$S_{yy} = \sum(y_i - \bar{y})^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n}$$

$$S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

Least-squares estimators:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\,\bar{x}$$

Regression line (fitted) & prediction:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x, \qquad \hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

Interpretation of coefficients: $\hat{\beta}_1$ = estimated change in Y per one-unit increase in X; $\hat{\beta}_0$ = predicted Y when $X = 0$.

Residuals: $e_i = y_i - \hat{y}_i$.

Sums of squares decomposition:

$$\text{SST} = \sum(y_i - \bar{y})^2 = S_{yy}$$

$$\text{SSR} = \sum(\hat{y}_i - \bar{y})^2 = \hat{\beta}_1\, S_{xy} = \frac{S_{xy}^2}{S_{xx}}$$

$$\text{SSE} = \sum(y_i - \hat{y}_i)^2 = \text{SST} - \text{SSR}$$

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}$$

Error variance estimator: $\hat{\sigma}^2 = \dfrac{\text{SSE}}{n-2}$.

Coefficient of determination:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \qquad 0 \le R^2 \le 1$$

Interpretation: proportion of variability in Y explained by the linear regression on X. Closer to 1 $\Rightarrow$ better fit.

Sample correlation coefficient:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2\sum(y_i - \bar{y})^2}}$$

Properties: $-1 \le r \le 1$; $R^2 = r^2$; $\text{sign}(r) = \text{sign}(\hat{\beta}_1)$.
Interpretation of r: $|r| \approx 1$ strong linear association; $r > 0$ positive, $r < 0$ negative; $r \approx 0$ no linear association (possibly nonlinear).

9. Quick Reference

Common z critical values:

$1-\alpha$	$z_{\alpha/2}$ (two-tail)	z_α (one-tail)
90%	1.645	1.282
95%	1.960	1.645
98%	2.326	2.054
99%	2.576	2.326

Common t critical values $t_{df,\alpha}$ (upper-tail):

$df \setminus \alpha$	0.10	0.05	0.025	0.01
5	1.476	2.015	2.571	3.365
10	1.372	1.812	2.228	2.764
15	1.341	1.753	2.131	2.602
20	1.325	1.725	2.086	2.528
25	1.316	1.708	2.060	2.485
29	1.311	1.699	2.045	2.462
30	1.310	1.697	2.042	2.457
40	1.303	1.684	2.021	2.423
60	1.296	1.671	2.000	2.390
∞	1.282	1.645	1.960	2.326

Use: two-tail $t_{df,\alpha/2}$ = column $\alpha/2$; e.g. 95% CI, $df = 20$: column 0.025 $\to$ 2.086. If df missing, round **down**.

Sample size for proportion (given margin E_0; round UP):
(a) $\hat{p}$ known:

$$n \ \ge\ \hat{p}(1-\hat{p})\left(\frac{z_{\alpha/2}}{E_0}\right)^2$$

(b) $\hat{p}$ unknown – conservative ($\hat{p} = 0.5$):

$$n \ \ge\ \frac{0.25\,z_{\alpha/2}^2}{E_0^2}$$

p-values – Two-Sample Z test ($H_0 : \mu_1 - \mu_2 = \Delta_0$):

H_1	p-value
$\mu_1 - \mu_2 \neq \Delta_0$	$2[1 - \Phi(\lvert Z_0 \rvert)]$
$\mu_1 - \mu_2 > \Delta_0$	$1 - \Phi(Z_0)$
$\mu_1 - \mu_2 < \Delta_0$	$\Phi(Z_0)$

10. Casio fx-580VN X – Quick Tips

Modes: `MENU` $\to$ **6** Statistics — **7** Distribution.
Stats (MENU 6): **1** 1-Var; **2** $y = a + bx$ linear reg.
Enter data $\to$ `AC` $\to$ `OPTN` for: $\bar{x}$, s_x ($= S$, sample SD), σ_x (pop SD), n, $\sum x$, $\sum x^2$. Regression adds $a = \hat{\beta}_0$, $b = \hat{\beta}_1$, r, $\hat{y}$, $\hat{x}$.
Always use s_x for S; square it for S^2.
Normal (MENU 7): **1** PD, **2** CD $P(a \le X \le b)$, **3** Inv Normal (returns x with $P(X \le x) = p$).
$P(Z > z)$: Normal CD, Lower$= z$, Upper$= 10$, $\mu = 0, \sigma = 1$.
$z_{\alpha/2}$: Inv Normal, Area$= 1 - \alpha/2$, $\mu = 0, \sigma = 1$.
t (MENU 7): **4** PD, **5** CD, **6** Inv t (lower-tail area).
$t_{n-1,\alpha/2}$: Inv t, Area$= 1 - \alpha/2$, $df = n - 1$.
Two-tail p-value: tCD(Lower$= |T_0|$,Upper$= 10$,df) $\times 2$.
Binomial (MENU 7): **A** PD $P(X = k)$, **B** CD $P(X \le k)$.
Type I & II Errors:

Decision	H_0 true	H_0 false
Fail to reject H_0	correct $(1-\alpha)$	Type II (β)
Reject H_0	**Type I** (α)	correct $(1-\beta)$

Power $= 1 - \beta = P(\text{reject } H_0 \mid H_0 \text{ false})$.
p-value rule: reject H_0 iff p-value $\le \alpha$.